

	This relationship is not required in the derivation of the expression.
16	C For full cycle processes, the net change in internal energy is zero. $\Delta U = Q + W$ $0 = Q + [(-4.2) + (1.0 \times 10^5 \times (20.0 - 5.0) \times 10^{-6})]$ $Q = 2.7 \text{ J}$
17	C